

Semiconductor Startup Landscape in India

An Investor's Perspective



SPECIALE INVEST



Founded in 2014, Ankur Capital is an early-stage venture capital firm in India, investing in digital and deep science technology companies.

Ankur Capital has backed over 30 companies across sectors.

This presentation was first presented at the 2nd Annual SemiX Summit at IIT Bombay in November 2024.

All data and information are as of November 2024.

Global Tailwinds for Semiconductors

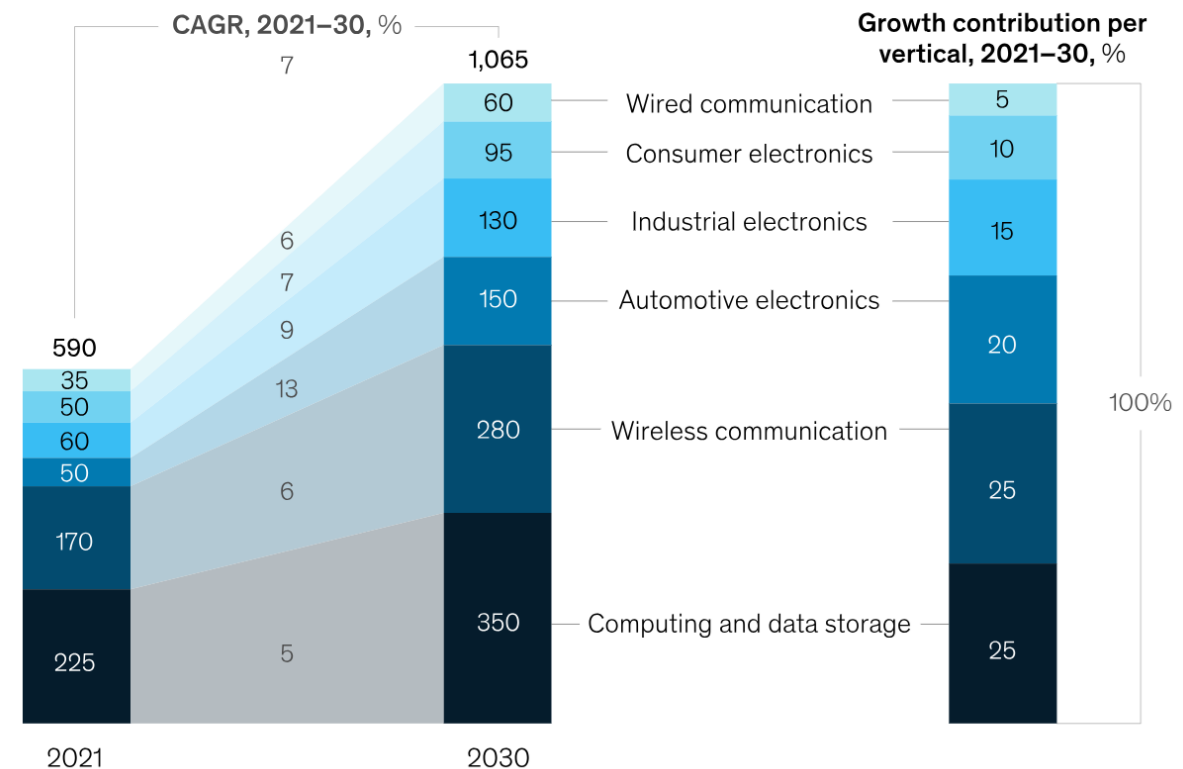
Ever Growing Demand: Semiconductor industry is already at \$600B and is growing at 15% per year.

New Age Applications coming up across IoT, Wireless, data centres, automotive, AI etc and pushing for new innovations and opportunities for startups.

Geopolitics / Supply Chain Interdependencies: re-shoring and de-centralizing of critical semiconductor resources supported by Govt incentives such as the CHIPS Act, ISM.

Competencies Cannot be Developed Overnight – either at a micro or a macro level; tech moats built in.

Global semiconductor market value by vertical, indicative, \$ billion



Source: McKinsey & Company, April 2022

India is on a Growth Trajectory

Year	Economy	Per Capita Income	Exports and Global Share	Human Capital
2023	~3.7 Trillion (5th Largest)	~2,600 USD (1.4 Bn)	700 Billion+ (~2.6%)	2nd largest workforce
2030	~7 Trillion (3rd Largest)	~4,000 USD (1.5 Bn)	2 Trillion (~6%)	Largest workforce
2047	~32 Trillion+ (2nd Largest)	~22,000 USD (1.6 Bn)	6 Trillion (~10%)	Among largest & youngest workforce in top economies

Source: Government of India

Stable Government at the Centre

Strong focus on Innovation and building an R&D ecosystem

After services, consumer and SaaS, we are truly in the **decade of building deep-tech hardware products**

Strong Capital Outlay Powering India as a Global Hub for Electronics

\$10B

Support for Semiconductor and Display Ecosystem

- Semiconductor Fabs and Display Fabs
- Compound Semiconductor and ATMP
- Design Linked Incentive (DLI)
- Modernization of Semiconductor Laboratory (SCL)

\$8.5B

Support for Electronics Manufacturing

- Production Linked Incentives for Mobile Phones, Components, IT Hardware
- Capex Linked Incentives for components, sub-assemblies
- Development of Electronics Manufacturing Clusters

\$13B

Support for Allied Sectors

Production Linked Incentives for:

- Advanced Cell Chemistries
- Automobiles and Auto Components
- Telecom and Networking
- Solar PV Modules
- Medical Devices

Diving into the \$10 Billion Outlay for Semiconductor Development

The 2024 Union Budget saw a 71% hike in funding for chip and electronics manufacturing.

Semicon Fabs

Fiscal Support: 50% of project cost on pari-passu basis

Technologies covered: All node sizes for Silicon CMOS based fabs

Display Fabs

Fiscal Support: 50% of project cost on pari-passu basis

Technologies covered: New-gen TFT LCD & AMOLED displays

Compound Semicon & Packaging

Fiscal Support: 50% of CAPEX pari-passu basis

Technologies covered: OSAT/ATMP, Compound Semiconductor, Silicon Photonics (including MEMS)

Semicon Design

Product Design Linked Incentive: 50% of eligible expenditure

Design Infrastructure Support

Renewed Focus on Semiconductor Startups in India

Government funding for R&D – DLI, C2S, along with tangential funding like TTDF, iDEX

Large emerging markets in India - Design in India, build from India for India – large markets in IoT, defence, vision, EVs which did not largely exist before.

Design in India, **build from India for the world** – Indian talent and startups building technologies for new-age applications data centres, Edge, AI etc

Talent availability – 20% of the global design talent is based out of India, **which has matured to be more holistic** – looked at tape outs, product, customer engagement, sales etc.



The infographic is titled 'Design Linked Incentive (DLI) Scheme' and is part of the 'Make in India Semiconductors for the World' initiative. It is issued by the Ministry of Electronics & Information Technology, Government of India, and features the myGov logo. The scheme is divided into three main sections: Fiscal Support, Tenure, and Beneficiaries. The Fiscal Support section details two types of incentives: Product Design Linked Incentive (up to 50% reimbursement) and Deployment Linked Incentive (6%-4% of net sales turnover). The Tenure section states a 5-year period starting from 01.01.2022. The Beneficiaries section lists 100 domestic semiconductor design companies. An illustration of a blue microchip is shown on the right side.

MINISTRY OF ELECTRONICS & INFORMATION TECHNOLOGY
GOVERNMENT OF INDIA

myGov
मेरी सरकार

Make in India
Semiconductors for the World

Design Linked Incentive (DLI) Scheme

Fiscal Support

Product Design Linked Incentive:
Reimbursement
Up to **50%**
of the eligible expenditure subject to a ceiling of ₹15 crore

Deployment Linked Incentive
Incentive of **6%-4%**
of net sales turnover over 5 years subject to a ceiling of ₹30 crore

Tenure

5 years
starting from 01.01.2022

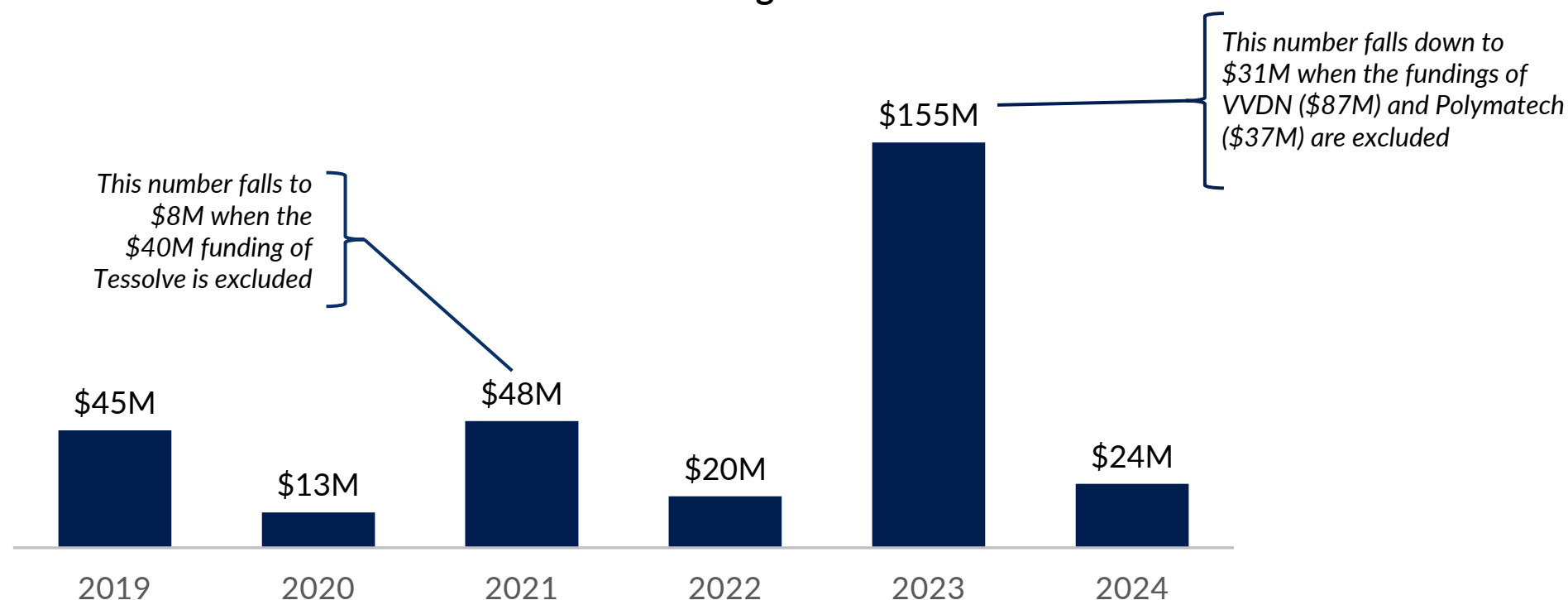
Beneficiaries

100
Domestic semiconductor design companies

India Semicon Funding Landscape

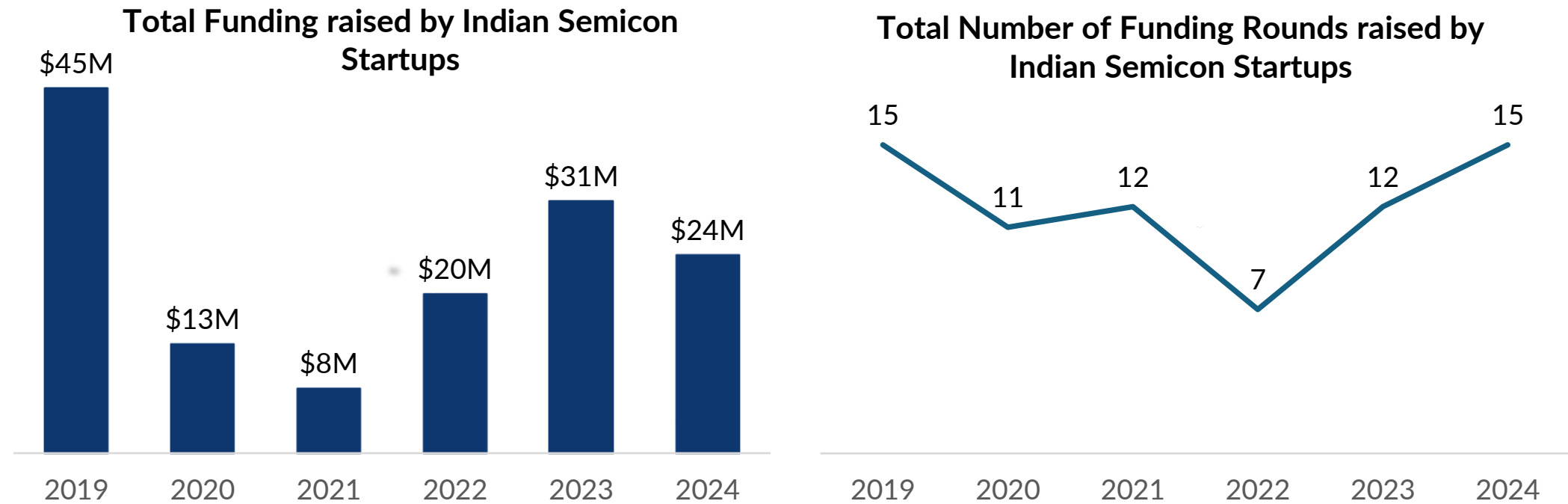
Over the past 5 years, \$300M has been invested in Indian semiconductor companies.

Total Semiconductor Funding in India



Source: Tracxn

India Semicon Funding has been Steadily Rising

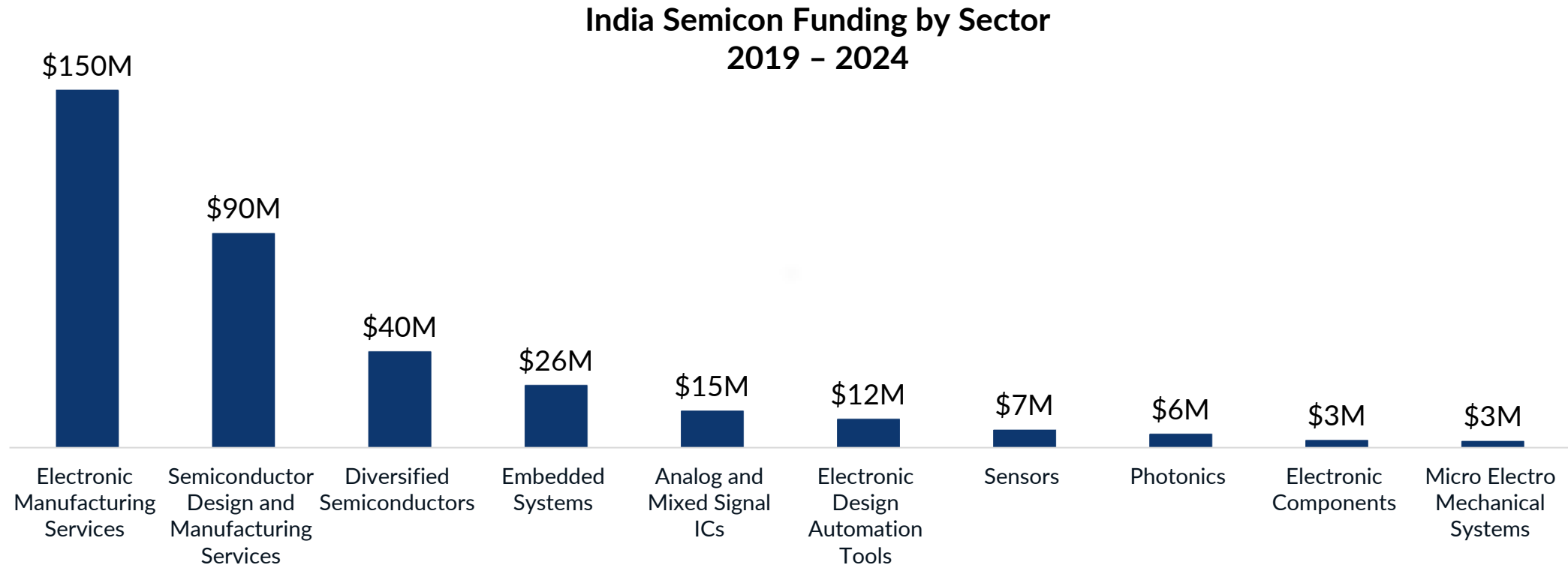


Source: Tracxn

Notable fundraises in 2024



Manufacturing and Design have seen the most Funding

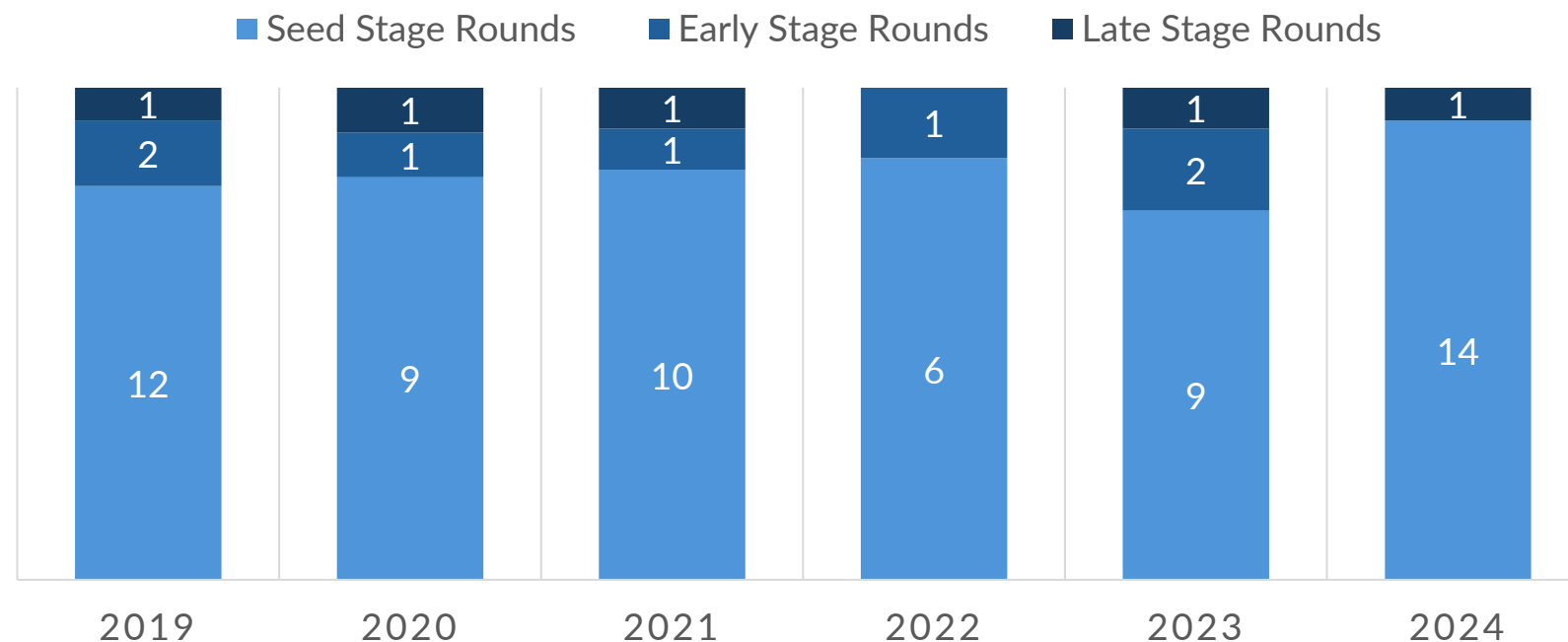


Source: Tracxn

70% of the funding has largely been in EMS and design services companies
However, we are seeing more money going into product startups over the past three years

Investors More Comfortable with Early Investments

India Semicon Funding Rounds by Stage



Source: Tracxn

We can expect a fair bit of follow-on rounds in the coming years in these startups who have raised seed rounds

Investor Landscape

Increasing Investor Appetite in Semiconductors – a lot of funds did their first Semicon investments over the past year

Financial VCs want to get in the journey early – targeting seed rounds

Corporate and Strategic investors have also shown an inclination to do this

VC	PE	Corporates	Angels
Speciale Invest	Motilal Oswal	Applied Materials	Ashish Kacholia
Peak XV	Zephyr Peacock	GAIL	
3one4 Capital	Novo Tellus	Shell	
Yali Capital		Bosch	
Pi Ventures		Cyient	
IIMA Ventures			
Vertex Ventures			

Startups have Achieved Notable Milestones Recently



Taped out their first **Secure IoT chip**



Unveiled **six-Core RISC-V test chip** to accelerate design adoption



Taped out a **POSIT-based silicon semiconductor chip**



Partnership with **Kaynes Semiconductors** for advanced packaging



Partnership with and funding from **Cyient**

DLI Beneficiaries



The Indian semiconductor ecosystem is shaping up to support the growth of product startups across verticals

Founder Profiles

Founders are coming out of large global design organizations to build products on their own – coming back with strong experience in design, product, customers, sales etc – Ex: OptoML, aiSIC, ABCRL, LineSemi, Makr



Professors and Researchers building startups based on research done in academia



Founders moving back to India from abroad looking to build in India.



Second-time founders in semiconductor space – moving from services to products



Ecosystems Supporting Semicon Startups

Semicon specific accelerators/CoEs
allowing for focussed support to semicon startups.



Private companies supporting
acceleration of semicon startups



Indian diaspora in the US offering
insights into the US market and getting
access to customers.



Corporate venture arms of semicon
majors are spending more time in India

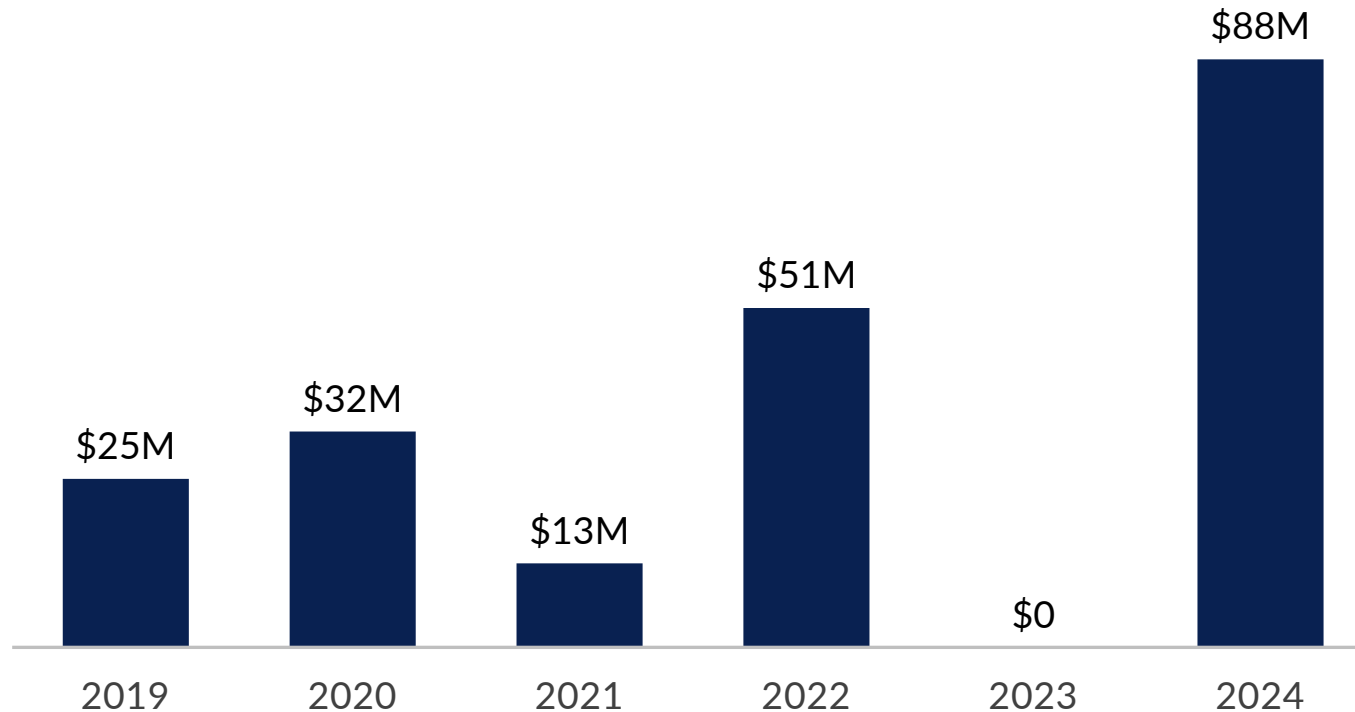


DLI/C2S are expected to follow a similar
path as BIRAC's BIG grants to provide
early support to semicon startups



Acquisitions in India Semicon

Indian Semicon Startups Total Acquisition Value



Source: Tracxn

Semiconductors make up an industry that has a rich track record of acquisitions

Past acquisitions of Indian companies give investors confidence that there light at the end of the tunnel

Acquirer Persona Taking Shape

Year	Company	Acquirer	Acquisition value (\$)
2024	Renesas' RF components unit	CG Power & Industrial Solutions	36M
2024	G.G.Trionics	CG Power & Industrial Solutions	38M
2024	SiliConch	L&T Semiconductor Technologies	16M
2024	Excelmax	Accenture	-
2024	Insemi	Infosys	33M
2022	P2F SEMI	Tessolve Semiconductor	-
2022	SteradianSemi	Renesas Electronics	-
2022	Saankhya Labs	Tejas Networks	37M
2020	Cerium Systems	Tech Mahindra	32M
2019	Sankalp Semi	HCL Technologies	25M

Whilst largely design services companies have been acquired by larger IT services companies, there is still a good track record of product companies such as Saankhya Labs, Steradian etc also being acquired.

The buyer persona, while evolving, is also becoming clearer.

Trends and Themes observed in 2024

Founders	Founders coming from large global design firms after 15+ years of work experience. Usually feel stagnated by working there and want to explore something more entrepreneurial. Academic research being productized through entrepreneurial professors
Funding	More startups getting access to DLI, C2S funding from Govt etc., Various profiles of VC investors now looking to invest in Indian Semicon startups
Products	Services companies now shifting to making products A large emphasis on building mass market chips for the Indian market - vision, IoT, smart meters, motor controllers etc. – but tech moat in these startups is not very evident.
Vision of Entrepreneurs	Building the next Texas Instruments / Qualcomm from India.
Customers	Indian customers showing a strong inclination for indigenous chips and technologies, often willing to co-develop the chips with startups (although they may not monetarily invest in the co-development process)
Partnerships and Acquisitions	Large IT services companies continue to acquire semicon IP design services companies; Gives confidence to investors about possible exit opportunities

Key Motivators for VCs to Invest in India Semicon

Favourable Geopolitics

Global geopolitical landscape propelling India to play a strong role in the Semicon ecosystem today

Strong Domestic Demand

Growing domestic demand across IoT, telecom, EVs, smart cities, 5G/6G, Defense etc. – each of them being large market opportunities in themselves

Build from India for the World

Global demand can be catered from Indian development

Local Talent

Strong local design talent and an academic setup that feeds this – makes it relatively easier for startups to hire talent

Govt Support

DLI/C2S funding gives more confidence to private investors to - underwrite the early-stage risk, validation from tech experts, extend runway

Track Record of Acquisitions

Both globally and in India, there is a decent track records of acquisitions. Gives confidence to investors to invest in this space

What Investors look for in Early-stage Semicon Startups

	Strong founding teams with domain expertise – usually with both hardware and software expertise.
Founders	CEOs/founders who have past background in engaging with customers and can handle business conversations
	Having a strong set of advisors to guide on product development, market access etc.
Customer conversations	Start conversing with potential customers early to understand what the requirements are. This is not to sell your product, but more to understand the pulse of the customer. Product development and business exploration should happen in parallel
	If your customers are in the US, prioritize spending time there at least 3-4 times a year.
Focus	While an IP/chip can solve for multiple use-cases, it would be good to have a narrow focus and understand where the beachhead market opportunity is and why that is the one to pursue

What Investors look for in Early-stage Semicon Startups

Grants and Partnerships

Partnerships that give you access to EDA tools and licenses

Leverage DLI, C2S and other grants that are available

Acquirability

The company should be able to create sufficient value in the value chain and thereby have visibility on who can be potential buyers 5-10 years down the line.

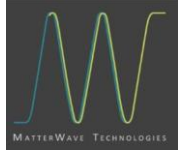
Fundraise

First fundraise should take you to an MPW at least (when applicable).

If you have successfully taped out a chip already, that adds to the investor confidence and potential valuation premiums.

Have a pragmatic view on what will get you your next fundraise

Areas of High Activity

Edge - IoT, Vision, Smart Meters etc.	Communication/Rf	Foundry Services	Photonics	Compound Semicon	Analog	Processors	Semicon Equipment
Rise in Edge applications – focus on reducing latency, reducing power consumption	Indigenous chips for defence and beyond	Make in India – cheaper and better	Faster compute, low power consumption	Applications beyond silicon	Increasing use of analog sensors, faster compute, low power consumption	Build from India for the world	Building an indigenous ecosystem
      	   	  	   	 	 	 	   

What We are Looking Forward to in 2025



Continued push from the Govt to encourage innovations in Semicon

We'll see **more founders from the large design companies** come out to build product startups

More seed rounds in Semicon startups

Funded companies hitting product milestones and raising their next round of funding

More investors will make their first Semicon investments as they build out their thesis in this space

Startups will start working with Indian OSAT-ATMP facilities for their requirements (Ex: Kaynes <> Lightspeed Photonics)

If you're building in this space, we'd love to have a conversation. Please reach out to vishal.katariya@ankurcapital.com.

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